



Digital Avionics Systems Conference
Montreal, Canada – September 14-18, 2025

Track: Urban, Advanced Air Mobility & New Space
Session: Airspace Design and Route Planning (AM5)

Investigating Cyclic Airspace Corridor Optimization for UAS Traffic Management based on Deep Reinforcement Learning

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AIRBUS

sopra  steria

Introduction

ATM

Standards & Regulations ✓
Experience ✓
Safety ✓

UTM

Technology ✓
Liberal Approach ✓

Efficiency ✗
Workload ✗
« Antifragile » ✗

Standards & Regulations ✗
Experience ✗



U-Space Airspace



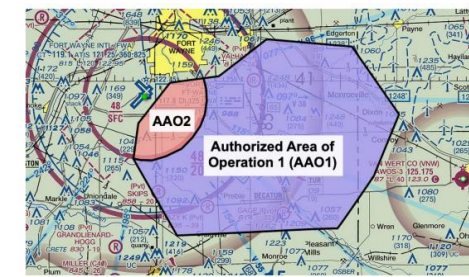
NAS extension



Airspace Design

Interface with^[2]:
Airspace Management
Geofencing
Mission Management
Conflict Management
Emergency Management
Environment
Interface with ATC

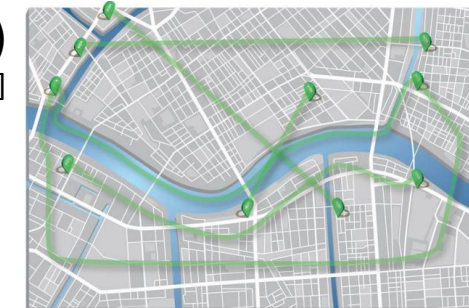
Performance Authorization^[1]



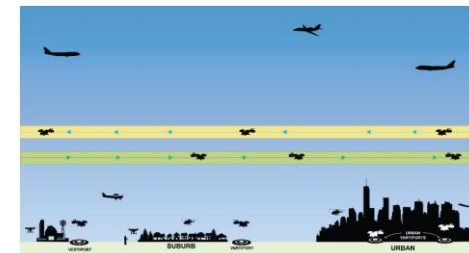
Mission Area^[3]



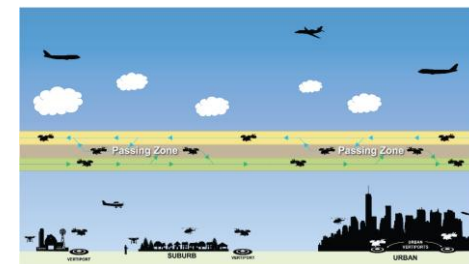
(Point-to-Point) Airspace Corridors^[1]



4D Operation Volume^[1]



Shared Airspace Corridor^[1]



Introduction

Cyclic Corridor Concept (C^3)

- Urban air traffic grows. Managing it manually is not scalable.
- C^3 introduces two unidirectional loops that pass by vertiports.
- All operators share the same airspace. Fair and simple.
- Less rerouting, fewer conflicts, and faster authorizations.

A Smart Way to Manage
Urban Traffic in the
Very Low Level?



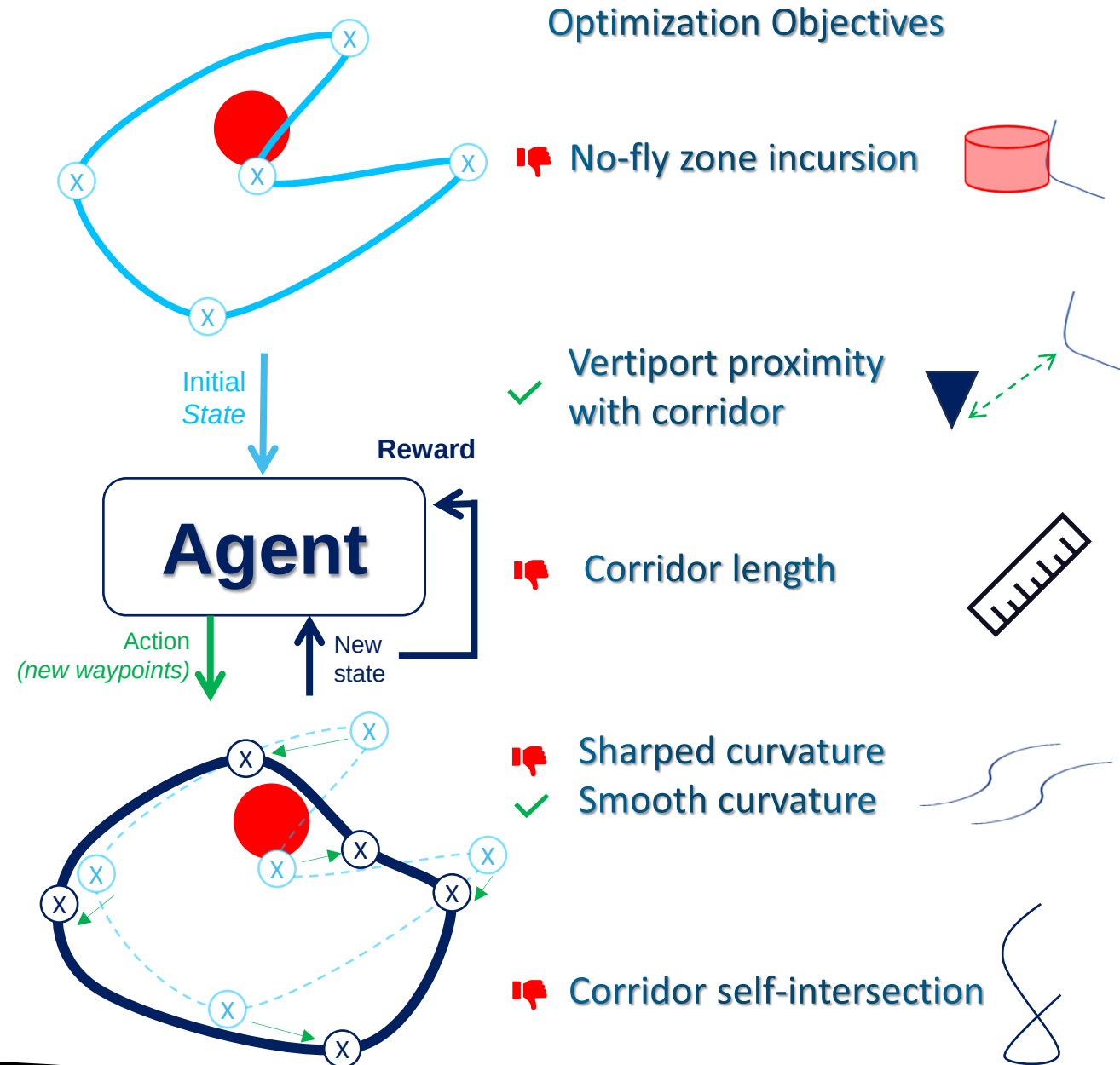
Introduction

We investigate the design of a cyclic airspace corridor in the hypothesis of a centralized (or federated but collaborative) strategic management of dense urban airspace at VLL.

- Deep Reinforcement Learning (DRL) models have been trained to generate holistic airspace corridors accounting for vertiports, obstacles and the scale of the airspace, in under one second.
- We compared our DRL approach with wider optimization strategies; gradient-based, local minimization algorithms as well as global optimizers.
- Spline interpolation is used to model realistic/feasible corridor geometries although aircraft performance models are not explicitly integrated to refine corridor curvature.
- To evaluate the design of the C3, it is benchmarked with Travelling Salesman problem and P2P flight planning.
- We validate our approach through simulated proof-of-concept use cases across three synthetic large-scale scenarios with quantitative flight efficiency metrics and qualitative safety assessment.

Methodology

- **Gamification for Design** : C^3 is reframed as a game-like optimization problem using Reinforcement Learning (RL).
- **AI-Driven Optimization**: The problem is formulated as a Markov Decision Process (MDP) to stepwise improve key waypoints by observing the environment and maximizing the reward.
- **Algorithmic Backbone**: Proximal Policy Optimization (PPO^[4]) and Soft Actor-Critic (SAC^[5]) RL algorithms drive the learning process,
- **Trajectory Modeling**: From the location of the key waypoints, the resulting airspace corridors are smoothly interpolated using Catmull-Rom^[6] splines.



Methodology

State Representation

We assign a fixed number n_c of control points $(C_i)_{i \leq n_c}$.

The state representation s of the corridor $\mathcal{C}^3$ is a composite vector containing three components:

I. $S_V^{(C_i)}$
7 elements



Closest vertiport
Farest vertiport

II. $S_0^{(C_i)}$
8 elements



Closest no-fly zone
Farest no-fly zone

III. S_c
23 elements

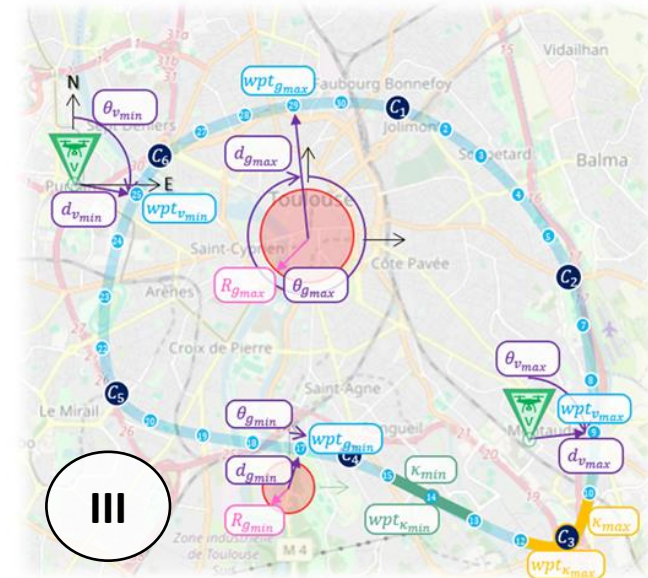
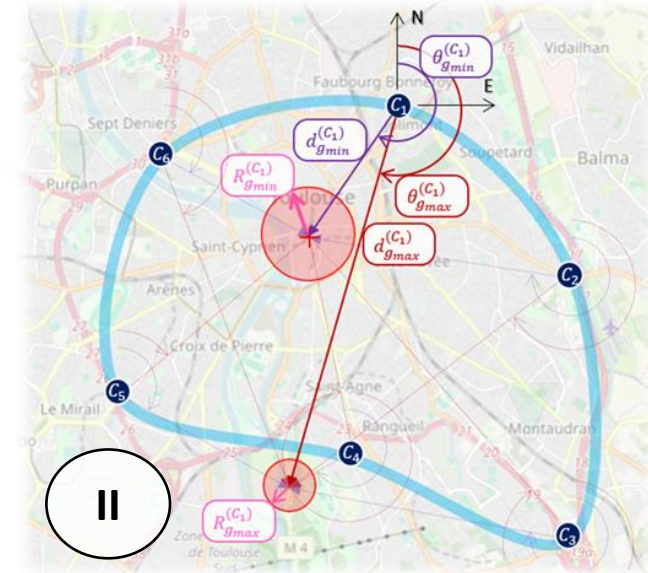
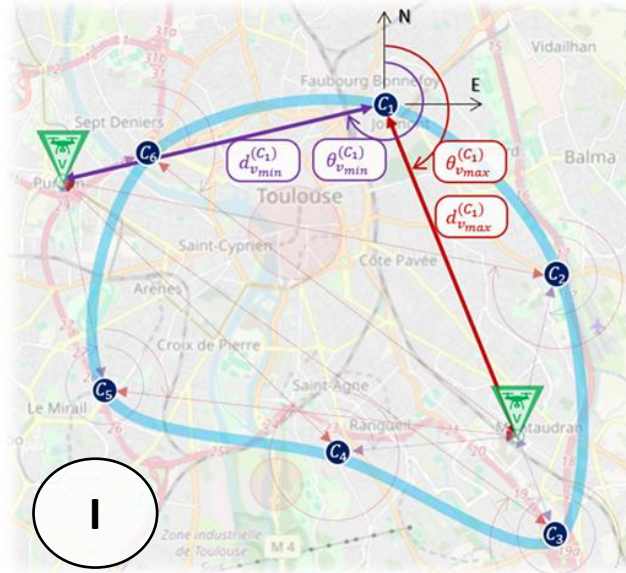
Entire corridor

$\mathcal{C}^3$

Closest vertiport and no-fly zone
Farest vertiport and no-fly zone

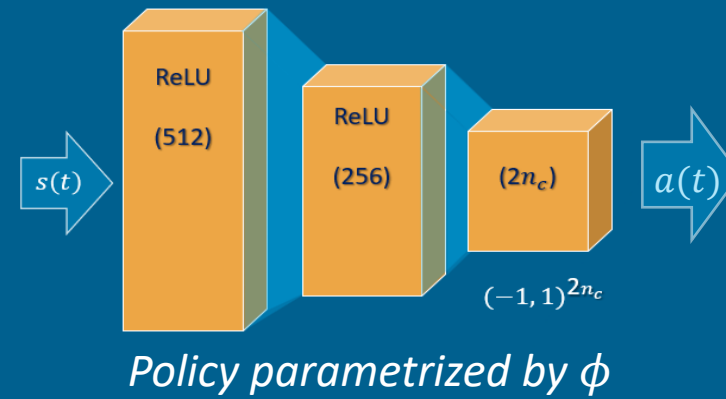
Environment

$$s(t) = \begin{bmatrix} S_V^{(C_1)} \\ \vdots \\ S_V^{(C_{n_c})} \\ S_0^{(C_1)} \\ \vdots \\ S_0^{(C_{n_c})} \\ S_c \end{bmatrix} (t)$$



Methodology

Action Space and State Transition



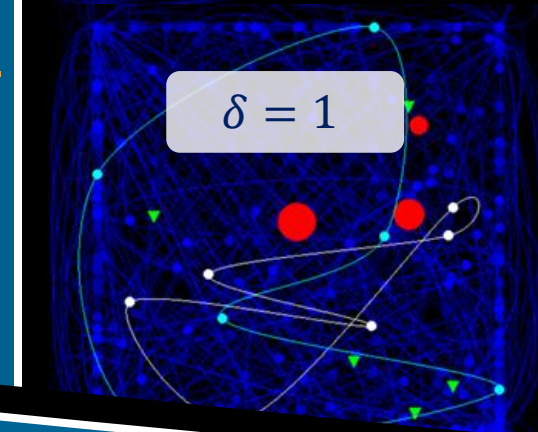
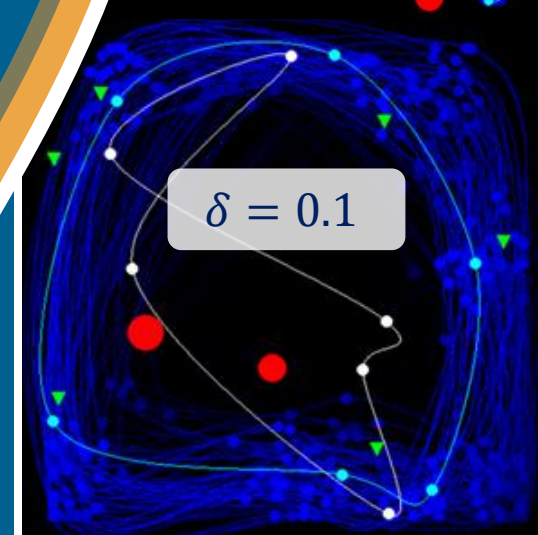
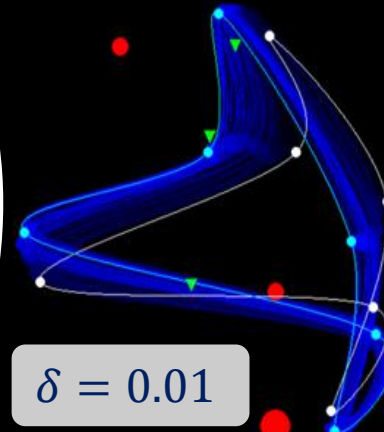
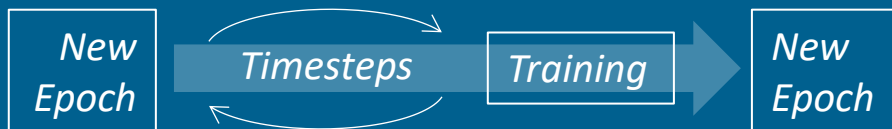
$$a(t) \sim \pi_{\phi}(\cdot | s(t)), \quad a \text{ shaped in } \begin{bmatrix} a_{\lambda}^{(c_1)} & a_{\lambda}^{(c_1)} \\ \vdots & \vdots \\ a_{\lambda}^{(c_{n_c})} & a_{\lambda}^{(c_{n_c})} \end{bmatrix}, \quad C = \begin{bmatrix} C_1 \\ \vdots \\ C_{n_c} \end{bmatrix}$$

$$C(t+1) = C(t) + \delta \cdot a(t), \quad \text{with } \delta \in (0, 1)$$

$\delta \rightarrow 0$: Exploration Stability

$\delta \rightarrow 1$: Exploration Stability

A small δ with more timesteps can compensate the lack of exploration but is slowing down the training.



- 100 timesteps
- Initial Corridor
 - Transition Corridor
 - Final Corridor
 - Vertiport
 - No-fly Zone

Methodology

Reward Function

No-fly zone incursion $r_g(t) = -\frac{1}{n_g} \sum_{1 \leq k \leq n_g} 1_{\{\min_{1 \leq i \leq n_c} (d_{gk}^{(c_i)}) < R_{gk}\}}$

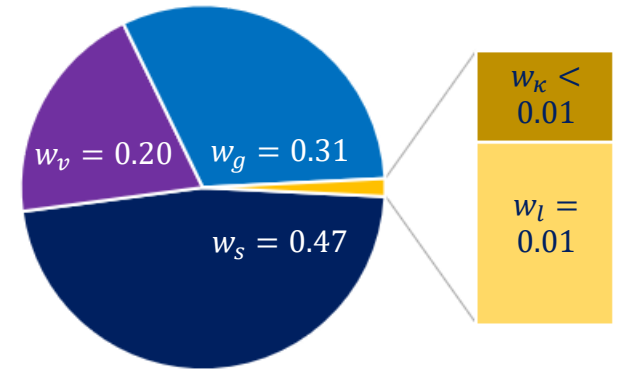
Vertiport Proximity $r_v(t) = \frac{1}{n_v} \sum_{1 \leq k \leq n_v} \text{clip} \left(\frac{d_{vmax} - d_{vk}^{(c_i)}}{d_{vmax} - d_{vmin}}, 0, 1 \right)$

Corridor length $r_l(t) = -\frac{d_c}{\max_{\mathcal{T} < t} (|r_l(\mathcal{T})|)}$

Corridor curvature smoothness $r_k(t) = -\frac{1}{\max_{\mathcal{T} < t} (|r_k(\mathcal{T})|)} \left(\sum_{1 < k \leq N} |\kappa_{wpt_k} - \kappa_{wpt_{k-1}}|^2 + |\kappa_{wpt_1} - \kappa_{wpt_N}|^2 \right)$

Corridor self-intersection $r_s(t) = -1_{\{\mathcal{C} \text{ self-intersect}\}}$

$$r(t) = w_g \cdot r_g(t) + w_v \cdot r_v(t) + w_l \cdot r_l(t) + w_k \cdot r_k(t) + w_s \cdot r_s(t)$$



Methodology

Training Parametrization

Parameters sim. env.

Parallel environments	96
Timesteps	100
Control points	5
Waypoints (include control points)	300
Vertiports	3 to 8
No-fly zones	1 to 8
No-fly zone radius	0.003 to 0.01



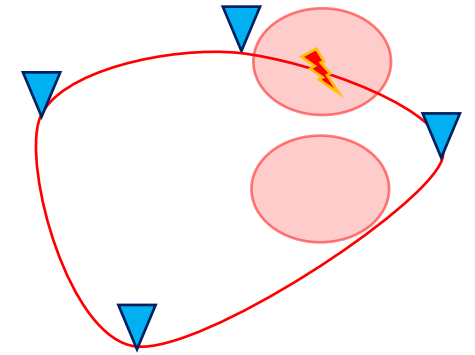
Parameters &
Presented data



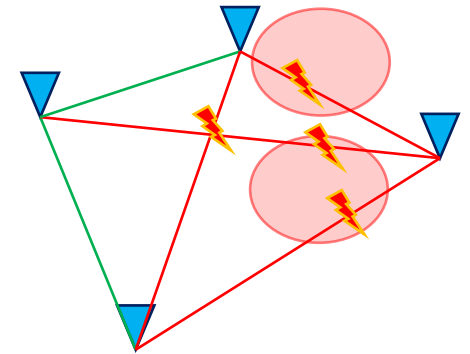
PPO & SAC
Algorithms

Methodology

Benchmarking



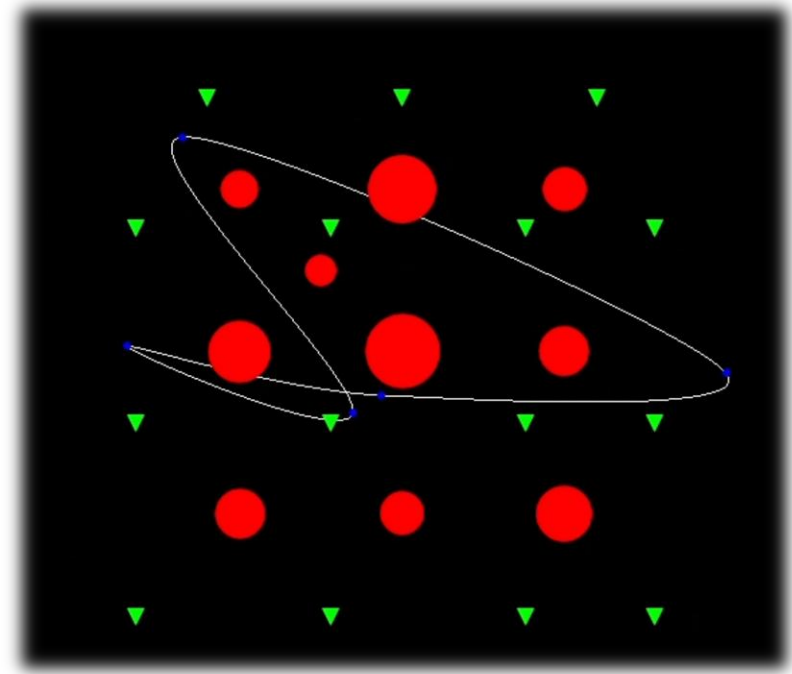
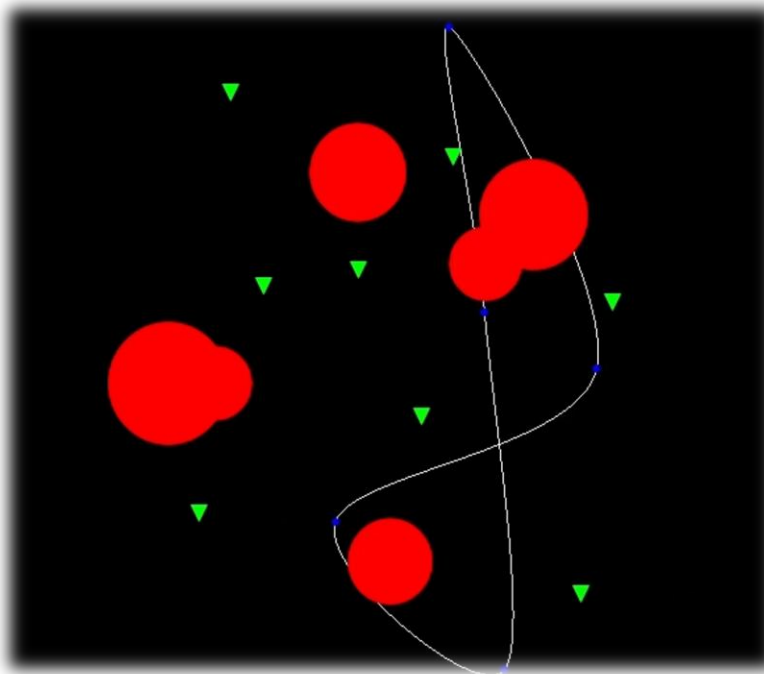
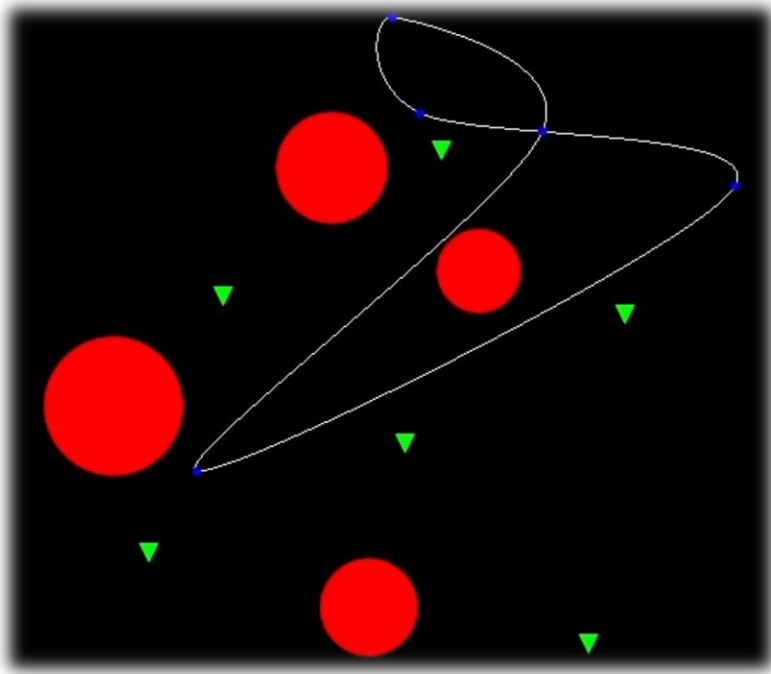
*Traveling Salesman Problem
(TSP)*



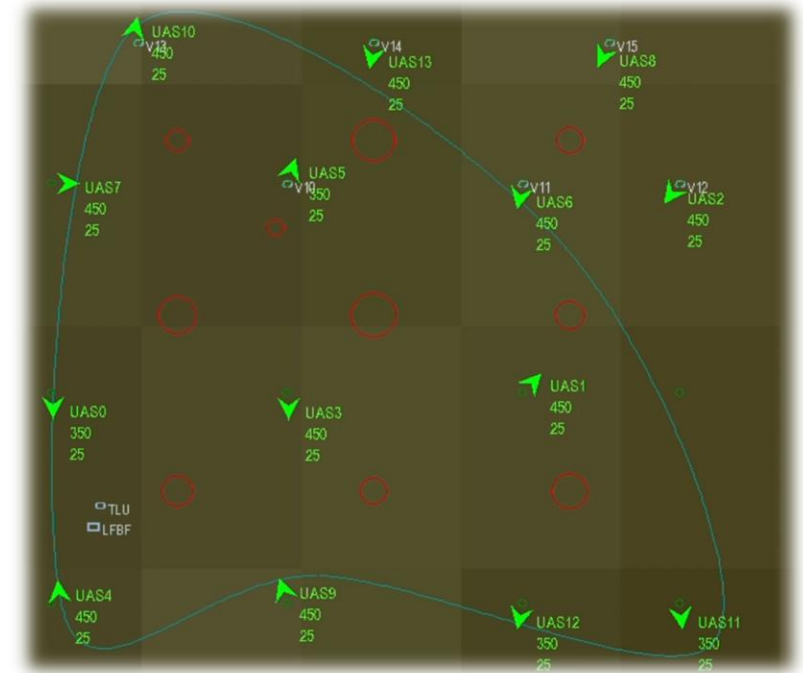
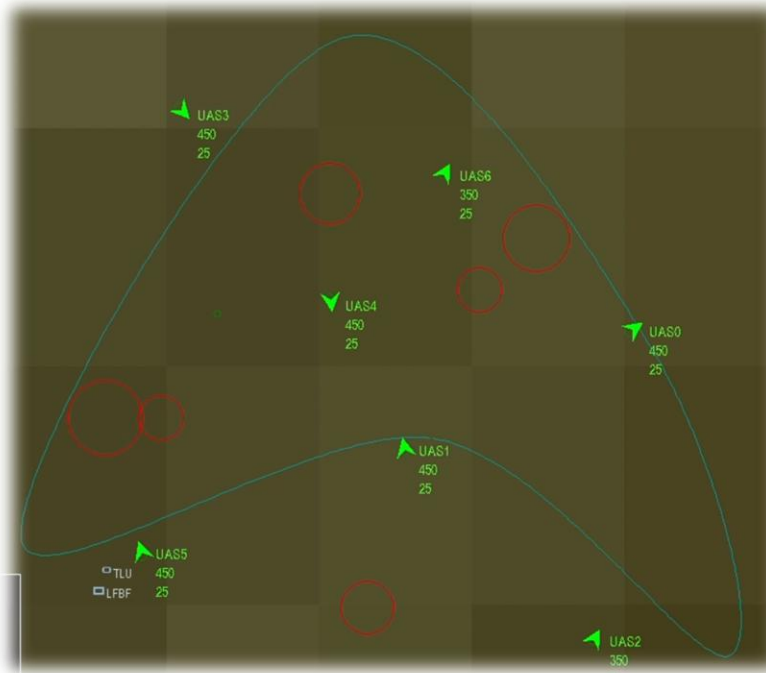
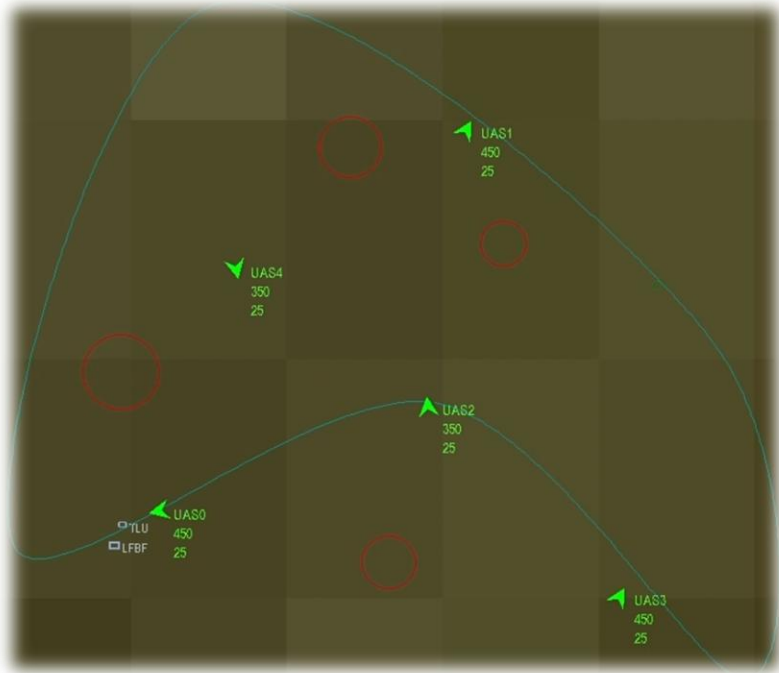
*Point-to-Point
(P2P)*

Experiments

Experiments



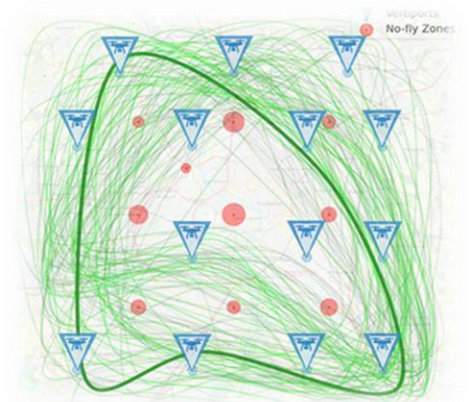
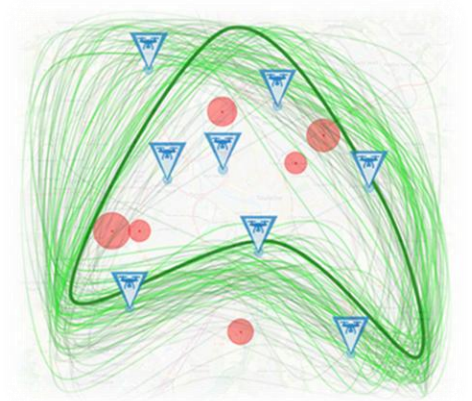
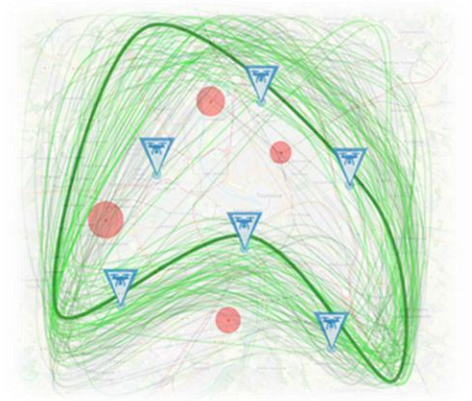
Experiments



Results

Scenario	I	II	III
Corridor Length	33.94 km	30.72km	46.09km
Doubled Extra-flight Distance	60.00%	50.00%	20.95%
Extra-flight Distance < 50%	26.67%	32.17%	52.38%
Prevented Flight Plan Intersections	11	46	1,127
Prevented No-fly zone Incursions	2	5	31
Prevented Rerouting	10	23	95
Total Routes (Vertiport Connections)	15	28	105

Results



Conclusion

1. Federal Aviation Administration, **Urban Air Mobility, Concept of Operations, Release, Version 2.0**, 2023.
2. SESAR Joint Undertaking, **U-space ConOps and architecture, CORUS, Edition 4**, 2023.
3. J.Zhao et al., **Cranfield AURD Frontend Simulation**, Part of High intensity Autonomous Drone Operations (HADO) project, UKRI Future Flight Challenges, 2024.
4. J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, **Proximal Policy Optimization Algorithms**, *arxiv 1707.06347* in *arXiv:1707.06347 [cs.LG]*, 2017
5. T. Haarnoja, A. Zhou, K. Hartikainen, G. Tucker, S. Ha et al., **Soft actor-critic algorithms and applications**, *arXiv preprint arXiv:1812.05905*, 2019.
6. C. Yuksel, S. Schaefer, and J. Keyser, **Parametrization and applications of Catmull-Rom curves**, *Computer-Aided Design*, vol. 43, no, 7, pp,747-755, 2011.



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